

Comparison of the relative merits of the midwave and longwave infrared bands for various target types using detected thermal contrast

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ABSTRACT

The two most important atmospheric transmission bands in the infrared occur at the midwave band of 3–5 μm and the longwave band of 8–12 μm respectively. For a given infrared detector a common question asked is, of the two, which, if any, gives the better performance? A question not without controversy, in this study an analysis designed to assess the relative merits of infrared detectors operating in either of these spectral bands based on the recently defined thermal figure of merit known as the detected thermal contrast is undertaken. Under ideal limiting conditions, by considering a range of targets whose spectral emissivities vary as a function of both wavelength and temperature, we impugn a number of results previously reported in the literature regarding the performance of a detector made using detected thermal contrast as the thermal figure of merit. For the two broad types of detectors considered it is found the midwave band for both thermal and quantum detectors gives better performance for a range of different target types.

Keywords: thermal contrast, spectral band comparison, detectors, emissivity, infrared, polylogarithms

1. INTRODUCTION

The two most important atmospheric transmission bands in the infrared occur at the midwave band of 3–5 μm and the longwave band of 8–12 μm respectively. When the question is asked which, if any, of the two bands leads to better detector performance, opinion continues to remain divided. A perennially asked question, the literature attests it is not without controversy. Conflicting and often contradictory results are found where conclusions often reflect the personal predilections of the proponent rather than being based on sound, quantifiable, technical factors. Manufacturers of infrared detectors only add to this already confused *mêlée* by suggesting the superiority of one band over the other, a decision based largely on the one they happen to have committed to manufacturing and selling! Indeed, for a given application many people coming to the infrared for the first time are somewhat surprised to learn the question of which spectral band gives better detector performance continues to remain open.¹ Our intention here is to provide guidance on this often vexed question using an analysis based on sound technical considerations free from any preconceived ideas or notions about which band we would like to see as being superior.

The question of detector performance was first given serious attention in the early 1970s by Jaeger, Nordbryhn and Stokseth.² They found for the detection of small temperature differences at ambient temperatures the longwave band in most cases was expected to be superior to the midwave band, while commenting that under certain conditions it was possible the latter could give equal to better performance compared to the former. Tuer,³ in a detailed analysis of the relative performance of midwave and longwave thermal-imaging systems operating in typical outdoor terrestrial settings showed that the midwave band gives superior performance compared to the longwave band in most situations. Performance however in the midwave band was found to deteriorate more rapidly compared to performance in the longwave band for cooler targets. Tuer concluded his analysis by stating that even though the performance in the midwave band may be slightly inferior compared to performance in the longwave band for most situations, this was not to say the former did not give adequate performance there. Renewed attention in the late 1980s^{4–7} saw a confused picture emerging with authors concluding by saying “well, it depends. . .” Later studies by Chrzanowski,⁸ Uçar and Özkan,⁹ and Sun and Zhu¹⁰ suggested the shorter waveband of 3–5 μm gave better performance while Chrzanowski even went as far as formulating this as a general rule.

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The approach used by each of the preceding authors assessed the performance of a detector from considerations based on calculations relating to signal-to-noise ratios (SNR). For small temperature differences between target and background, using

$$\text{SNR} = \frac{i_s}{i_n}, \quad (1)$$

where i_s is the signal current while i_n is the noise current, it is possible to calculate the signal-to-noise ratio which gives a reasonable measure of the measured signal detected by an infrared detector. The problem is such a calculation is very susceptible to errors made in the assumptions used, particularly in the emissivity of the target, and is the principal reason why a definitive answer concerning which spectral band leads to greater detector performance has not been reached.

More recently, to help overcome the inherent uncertainties associated with attempting to quantify performance using calculations based on SNR, Páez and Strojnik proposed what has turned out to be a more robust measure of detector performance in the form of a new thermal figure of merit known as the *detected thermal contrast*.¹¹ As a concept in infrared imaging, thermal contrast provides an important measure of performance as it allows a target to be distinguished from its background on the basis of temperature alone. The detected thermal contrast attempts to quantify the overall performance of an infrared detector from the initial emission of thermal radiation at the surface of a target to the final measurable input signal seen in the detecting instrument based on changes in temperature occurring at the surface of the target.

Since its introduction a number of analyses assessing the performance of detectors in the infrared using detected thermal contrast have been undertaken.^{12–25} Initially each was made using a number of different approximation schemes with greatly conflicting results reported. In the work by Páez *et al.*^{13–19,23,24} the detected thermal contrast was calculated using a truncated series approximation^{26–28} they termed a *compensated approximation** for targets whose emissivities depended only linearly on temperature. For the two types of detectors considered, these being *thermal* (relying on a heating effect for detection) and *quantum* (one which essentially counts photons utilizing the direct interaction between the incident photons and the electrons of the detector), they found for either detector type the detected thermal contrast is higher in the longwave band compared to the midwave band. Páez *et al.* also reported the size of the detected thermal contrast is significantly larger for a quantum detector compared to a thermal detector, exhibiting up to 20 times better performance over the latter. Fang²¹ obtained results for the detected thermal contrast using a Gauss–Laguerre approximation^{29,30} to the infinite series appearing in his expressions but provided no statement regarding detector performance.

Work performed by the author so far^{25,31,32} found, using an exact formulation based on the so-called polylogarithm function,³³ for targets whose emissivities depend only linearly on temperature the midwave band gave the larger value for the detected thermal contrast, and hence the better performance. Values for the detected thermal contrast within either band for thermal and quantum detectors were also found to be very similar suggesting the performance by detector type is not as an important factor as Páez *et al.* seem to suggest. So one can see the question of which of the two principal atmospheric windows in the infrared leads to better performance in a detector is far from settled. Even when similar formulations using the simplest of models have been used, conflicting and contradictory results remain. Part of the aim of the present work is to help better clarify each of these conflicting results.

In this paper we apply the results of our earlier work^{25,32} to a range of different target types for the general case where the spectral emissivity of the target is taken to be a linear function of the wavelength, temperature, and wavelength–temperature product. The treatment applies to focal-plane thermal-imaging systems used for the passive detection of radiation from terrestrial sources mainly utilized in short range applications. To facilitate the analytic treatment a mathematical approach based on the recently introduced polylogarithmic formulation developed by the author is used.^{34,35} For the range of target types considered the present work impugns several results previously reported in the literature relating to the behaviour of the detected thermal contrast for a given spectral interval and detector type. In line with our previous findings, we show the midwave band for either detector type leads to better performance while differences between detector type and the contribution arising from considering target spectral emissivities which are functions of both wavelength and temperature are found to be not as significant as one might initially expect.

*Their approximation was compensated in the sense the number of terms used in the truncated series is adjusted according to the size of an error term with the error term being included in the series summation.

2. MATHEMATICAL FORMULATION

To help motivate the definition used for detected thermal contrast, let us start by considering the thermal contrast for a blackbody. Thermal contrast allows the distinction between an object and its background to be made solely on the basis of its temperature. For a blackbody source the spectral thermal contrast $\mathcal{T}_{\lambda}^b$ corresponds to the fractional change in the spectral radiant exitance $M_{e,\lambda}^b(\lambda, T)$ over the fractional change in temperature T of the body. Mathematically this can be expressed as

$$\mathcal{T}_{e,\lambda}^b = \frac{dM_{e,\lambda}^b}{M_{e,\lambda}^b} \bigg/ \frac{dT}{T} = \frac{T}{M_{e,\lambda}^b} \left(\frac{dM_{e,\lambda}^b}{dT} \right). \quad (2)$$

As is customary, the subscript λ for wavelength indicates a spectral quantity is being considered, the prefix “e” indicates the radiometric case[†] has been selected while the superscript “b” is used on account of the body under consideration is that of a black body. It is a positive, dimensionless quantity.

Spectral quantities, being differential in nature, have no direct physical meaning. Rather, only those quantities defined on a finite spectral interval which are measurable have real physical meaning.³⁷ On moving to a finite spectral band $[\lambda_1, \lambda_2]$ ($\lambda_2 > \lambda_1$) the spectral radiant exitance (a differential quantity) is replaced by an equivalent quantity within a finite spectral band of

$$M_{e,\lambda_1 \rightarrow \lambda_2}^b = \int_{\lambda_1}^{\lambda_2} M_{e,\lambda}^b(\lambda, T) d\lambda. \quad (3)$$

The above integral is well known. It often appears in problems related to the study of radiative heat transfer with its canonical dimensionless form being expressed in terms of the equally well-known *blackbody fractional function of the first kind*.^{38,39} Within a finite spectral band, after suitably modifying the spectral form (that is, Eq. (2)) to its in-band form, the thermal contrast for a blackbody can be expressed as

$$\mathcal{T}_{e,\lambda_1 \rightarrow \lambda_2}^b = \frac{T \frac{d}{dT} \int_{\lambda_1}^{\lambda_2} M_{e,\lambda}^b(\lambda, T) d\lambda}{\int_{\lambda_1}^{\lambda_2} M_{e,\lambda}^b(\lambda, T) d\lambda}. \quad (4)$$

Taking the idea of thermal contrast for a blackbody (namely, Eq. (4)) and extending it to real targets, the thermal contrast detected by either thermal or quantum detectors for focal-plane radiation detecting instruments can be expressed as

$$\mathcal{T}_{s,\lambda_1 \rightarrow \lambda_2}^D = \frac{T \frac{d}{dT} \int_{\lambda_1}^{\lambda_2} \mathfrak{D}_{s,\lambda}^D(\lambda, T) d\lambda}{\int_{\lambda_1}^{\lambda_2} \mathfrak{D}_{s,\lambda}^D(\lambda, T) d\lambda}. \quad (5)$$

Here T is the absolute temperature at the surface of the target while $\mathfrak{D}_{s,\lambda}^D(\lambda, T)$ is the effective spectral irradiance as detected and measured by the sensor in the system. As defined, the detected thermal contrast is dimensionless. The superscript “D” is used to indicate quantities relating to the detector.

The effective spectral irradiance at the detector is determined by the actual spectral irradiance $\mathfrak{D}_{s,\lambda}(\lambda, T)$ at the detector weighted by a (dimensionless) spectral response of the detector $\mathcal{R}_s(\lambda)$. It can be expressed as

$$\mathfrak{D}_{s,\lambda}^D(\lambda, T) = \mathcal{R}_s(\lambda) \mathfrak{D}_{s,\lambda}(\lambda, T) = \Omega \tau(\lambda) \mathcal{R}_s(\lambda) L_{e,\lambda}(\lambda, T) = \Omega \tau(\lambda) \mathcal{R}_s(\lambda) \varepsilon_\lambda(\lambda, T) L_{e,\lambda}^b(\lambda, T). \quad (6)$$

Here Ω is a solid angle subtended by the optical system at the detector plane, $\tau(\lambda)$ is the wavelength dependent transmission loss factor, $\mathcal{R}_s(\lambda)$ a dimensionless weighted spectral response of the detector, $\varepsilon_\lambda(\lambda, T)$ the spectral emissivity for the target

[†]Radiometric quantities are those quantities used to describe radiant energy and are based on the joule (or watt). It is the natural unit of choice for thermal detectors and represents the so-called “energetic” system of units. Contrast this to actinometric quantities which are those quantities related to the measure of photons and are more appropriate for quantum detectors. They represent the so-called “photonic” system of units.³⁶

while $L_{e,\lambda}^b(\lambda, T)$ is the spectral radiance for the equivalent blackbody source. The prefix ς is a generalised subscript. It is introduced to differentiate between the radiometric case ($\varsigma = e$) corresponding to thermal detectors and the actinometric case ($\varsigma = q$) corresponding to quantum detectors. Different forms for $\mathcal{R}_\varsigma(\lambda)$ allow for the two different types of detectors to be handled and depends on the underlying physics of how the incident radiation is detected.

3. MODEL SYSTEM

3.1 Approximations and assumptions

In modelling typical operating conditions a number of assumptions and simplifying approximations are made. Firstly, the present analysis is limited to stable atmospheric conditions. Distances between the target and detector are assumed to be reasonably short. Both these assumptions are representative of conditions usually found in most industrial and scientific settings requiring accurate measurement of temperature distributions over the surface of a target. Under these assumptions the transmission loss factor can be taken to be independent of wavelength, that is, $\tau(\lambda) \simeq \tau$.

The form for the spectral emissivity of the target is generally not known in advance, and even when it is, is difficult to determine exactly. However, for well studied targets such as tungsten, copper, nickel, platinum, or gold, over narrow wavelength and temperature ranges, it is well modelled by a linear function of its wavelength and temperature.^{40–43} We therefore model the target spectral emissivity within each infrared atmospheric transmission band empirically using an equation of the form

$$\varepsilon_\lambda(\lambda, T) = a_0 + a_1 T + a_2 \lambda + a_3 \lambda T. \quad (7)$$

Here a_0, a_1, a_2 , and a_3 are constants and depend on the target material. Each of the constants is found by fitting published experimental spectral emissivity data to Eq. (7) in each spectral interval of interest using multiple linear regression.

Finally, the well-known approximations for the weighted spectral response applicable to ideal detectors operating in the infrared will be employed.⁴⁴ In the case of a thermal detector one has $\mathcal{R}_e(\lambda) = k$, $0 < k < 1$, while for a quantum detector⁴⁵

$$\mathcal{R}_q(\lambda) = \begin{cases} \frac{\lambda}{\lambda_{\max}} & : 0 < \lambda \leq \lambda_{\max} \\ 0 & : \lambda > \lambda_{\max} \end{cases} \quad (8)$$

Here $\lambda_{\max}$ is some fixed upper wavelength for which a response in the detector is still possible and will be assumed to be beyond the upper limit for the longwave band of 12 μm .

With the above assumptions and approximations made, expressions for the detected thermal contrast for both thermal and quantum detectors can be found. They can be expressed in closed form in terms of one of the special functions of mathematical physics known as the polylogarithm function.[‡] Details leading to expressions for the thermal contrast for either detector type are not given here but instead can be found elsewhere in previous work of the author.^{25,32}

3.2 Spectral emissivities

The following four target types are considered: tungsten (W), nickel (Ni), copper (Cu), and the titanium alloy Ti-6Al-4V.[§] These four materials have been chosen as values for spectral emissivities at relatively low temperatures (< 1000 K) and at wavelengths in the midwave and longwave spectral bands are relatively easy to come by in the literature. For each type of material, data reported in the literature for spectral emissivities in the midwave and longwave bands at relatively low temperatures is selected and fitted to Eq. (7) using multiple linear regression.

For tungsten, for wavelengths above 1.5 μm , qualitatively its spectral emissivity decreases with increasing wavelength, while for fixed wavelength, decreases as the temperature decreases (see Figure 234A on page 796 of Touloukian and DeWitt⁴⁶). Spectral emissivity data reported by Aksyutov⁴⁷ will be used for tungsten. For nickel and copper, in general

[‡]The polylogarithm function, or polylogarithms for short, is one of the classical special functions of mathematical physics known for over 250 years. While known for many years, only in the last few decades has one seen a sudden resurgence in interest and use of the function. The function can be thought of as a natural generalisation of the logarithmic function. For an account of these functions the book by Lewin is highly recommended.³³

[§]Ti-6Al-4V is a titanium alloy consisting of 6% aluminium and 4% vanadium. It is widely used in many engineering and biomedical applications due to its excellent combination of high strength, low density, and high resistance to corrosion.

Table 1: Spectral emissivity coefficients for each of the four targets obtained by empirically fitting spectral emissivity data to $\varepsilon(\lambda, T) = a_0 + a_1T + a_2\lambda + a_3\lambda T$ using the method of linear multiple regression. In the expression for $\varepsilon(\lambda, T)$ temperature is measured in Kelvin while wavelength is measured in micrometres.

Target		a_0	a_1 [K ⁻¹]	a_2 [μm ⁻¹]	a_3 [K ⁻¹ μm ⁻¹]	Temperature [K]
W	3–5 μm	1.178×10^{-2}	1.485×10^{-4}	-1.159×10^{-4}	-1.869×10^{-5}	$273 \leq T \leq 873$
	8–12 μm	9.320×10^{-3}	6.457×10^{-5}	-2.271×10^{-5}	-2.429×10^{-6}	
Ni	3–5 μm	1.495×10^{-1}	9.358×10^{-5}	-1.370×10^{-2}	-6.514×10^{-6}	$539 \leq T \leq 946$
	8–12 μm	8.547×10^{-2}	7.414×10^{-5}	-3.257×10^{-3}	-9.881×10^{-7}	
Cu	3–5 μm	4.974×10^{-2}	1.434×10^{-5}	-6.374×10^{-3}	6.318×10^{-7}	$526 \leq T \leq 908$
	8–12 μm	2.853×10^{-2}	6.692×10^{-6}	-1.715×10^{-3}	1.366×10^{-6}	
Ti-6Al-4V	3–5 μm	4.858×10^{-1}	-1.523×10^{-4}	-3.108×10^{-2}	8.727×10^{-6}	$546 \leq T \leq 949$
	8–12 μm	3.946×10^{-1}	-1.663×10^{-4}	-1.252×10^{-2}	6.994×10^{-6}	

their respective spectral emissivities decrease with increasing wavelength and decreasing temperatures (see Figure 133 on page 434 for nickel and Figure 48 on page 153 for copper of Touloukian and DeWitt⁴⁶ respectively). Spectral emissivity data recently reported by Setién-Fernández, Echánit, González-Fernández, Pérez-Sáez, and Tello,⁴³ for both nickel and copper is used. For Ti-6Al-4V relatively few studies into the determination of its spectral emissivity as a function of wavelength and temperature have been undertaken. The recent data reported by González-Fernández, Risueño, Pérez-Sáez, and Tello⁴⁸ to fill this gap seems appropriate and will accordingly be used. Values for the four spectral emissivity coefficients a_0 through to a_3 obtained by fitting published spectral emissivity data to Eq. (7) in the midwave and longwave bands using multiple linear regression for each of the four different target types considered are given in Table 1.

4. RESULTS AND DISCUSSION

In Fig. 1 the detected thermal contrast as a function of temperature for the two most common atmospheric windows in the infrared for four different targets whose spectral emissivities have been modelled using a linear function for the wavelength, temperature, and wavelength–temperature product is shown. The graph to the left is for a thermal detector while that to the right is for a quantum detector.

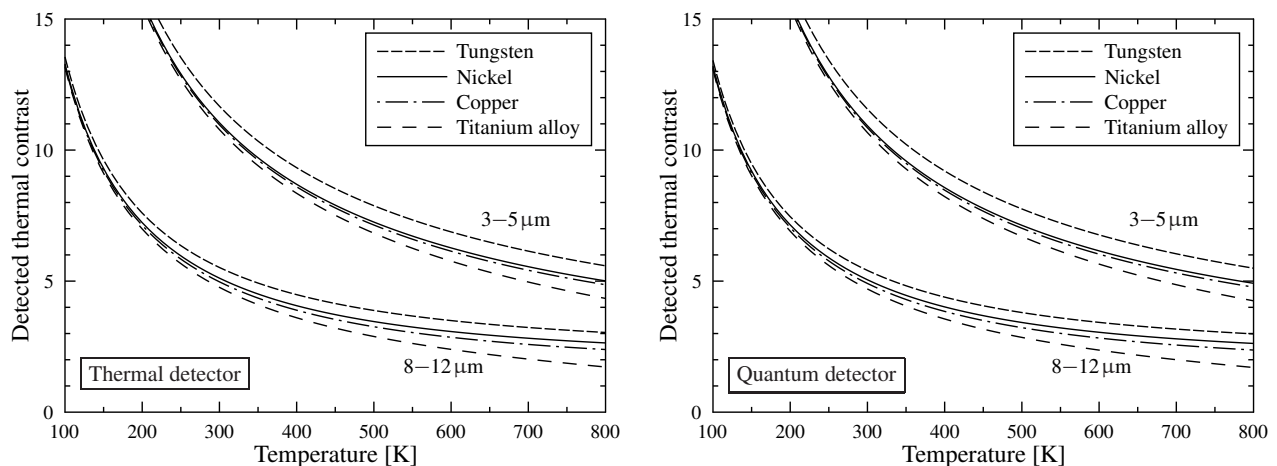


Figure 1: Detected thermal contrast as a function of the temperature for four different target types at the midwave (top four curves) and longwave (bottom four curves) bands respectively. The different target types considered are: tungsten (small broken line), nickel (solid line), copper (dot dashed line), and the titanium alloy Ti-6Al-4V (large broken line). LEFT: Thermal detector. RIGHT: Quantum detector.

For target types whose spectral emissivities are well modelled by a linear spectral emissivity in wavelength and temperature it can be seen that for either detector type the midwave band gives the larger value for the detected thermal contrast. The difference in detected thermal contrast is more pronounced at temperatures close to room temperatures with the difference gradually becoming smaller as the temperature of the target increases. This is to be expected as small changes in temperature for targets at elevated temperatures provide less contrast compared to temperature changes occurring at the surface of the target at much lower temperatures. Furthermore, no discernible difference in the performance of thermal and quantum detectors for each target type is observed.

Within the same spectral band the results between different target types are similar. This is to be expected, since as the author has previously shown,³² detected thermal contrast is a very robust measure of performance, particularly when it comes to the many inherent uncertainties associated with target spectral emissivities. Small changes in the spectral emissivity for each target does not lead to significant changes in the value found for the detected thermal contrast.

Reconfirming our earlier work,^{25,32} the results reported here continue to remain at odds with those reported by Aranda, Pérez and Scholl.²⁴ The latter found the detected thermal contrast for either detector type is higher in the longwave band compared to the midwave band. They also reported the size of the detected thermal contrast is significantly larger for a quantum detector compared to a thermal detector, exhibiting up to 20 times better performance for the former compared to the latter. We, on the other hand, continue to not observe this. Instead, for the range of target types considered within the linearly wavelength and temperature dependent spectral emissivity model, performance between thermal and quantum detectors are equally matched in line with what has been reported for infrared detectors elsewhere.⁴⁹

5. CONCLUSION

Using an analysis based on detected thermal contrast the relative merits of thermal and quantum based detectors operating in the midwave and longwave infrared bands have been assessed. For focal-plane thermal-imaging systems used for the passive detection of radiation from terrestrial sources in stable atmospheric conditions over short ranges whose spectral emissivities are linearly dependent on both wavelength and temperature we have shown the midwave band gives the better performance while differences between detector type plays no significant role in performance. Examples were given for targets made of tungsten, nickel, copper, and the titanium alloy Ti-6Al-4V but the analysis developed applies equally to any target type whose spectral emissivity is well modelled by a function which is linearly dependent on wavelength and temperature. Results of the present work are based on the polylogarithmic formation recently developed by the author. It extends a number of approximate schemes that have been developed in the past where target spectral emissivities linearly dependent only on temperature have been considered. Finally, we hope the forgoing analysis has allowed a reasonable answer to the question of detector performance to be at last reached.

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